

5

GCR equation derivation EoR-FG: the systematics as gain case

Glossary of terms:

e: EoR component (a vector)

a_{fg} : Foreground component (a vector)

b_{sys} : Systematics component (a vector)

n: noise

B: The systematics covariance (defined as $B = \langle b_{sys} b_{sys}^\dagger \rangle$)

E: EoR covariance ($E = \langle e e^\dagger \rangle$)

G: Foregrounds covariance ($G = \langle a_{fg} a_{fg}^\dagger \rangle$)

N: Noise covariance matrix

H: The projection operator

Maths work-through:

Let us define $y = \mathbf{H} @ b_{sys}$ as the systematics component. If we treat this component as the gain and the "signal" component as:

$$s = (e + \mathbf{G} a_{fg})$$

the complete data model (with noise) is defined as:

$$d = y(e + \mathbf{G} a_{fg}) + n$$

Now the Bayesian posterior is defined as:

$$p(s|S, N, d) \propto p(d|s, S, N) p(s|S)$$

After the Gaussian approximation, this becomes:

$$p(s|S, N, d) \propto \exp[-(d - s)^\dagger N^{-1} (d - s)] \exp[-s^\dagger S^{-1} s]$$

Expanding the prior, taking the log of the posterior, we can find the maxima of the above by doing:

$$\frac{\delta}{\delta e} \left[-(d - s)^\dagger N^{-1} (d - s) - e^\dagger y^\dagger E^{-1} y e - a_{fg}^\dagger y^\dagger G^{-1} y a_{fg} \right] = 0$$

and

$$\frac{\delta}{\delta a_{fg}} \left[-(d - s)^\dagger N^{-1} (d - s) - e^\dagger E^{-1} e - a_{fg}^\dagger G^{-1} a_{fg} \right] = 0$$

Expanding s to $s = y(e + a_{fg})$ and performing the above equations, we are left with this system of two equations:

1. $(y^\dagger N^{-1}y + y^\dagger E^{-1}y) e + (y^\dagger N^{-1}yG) a_{fg} = y^\dagger N^{-1}d$
2. $(G^\dagger y^\dagger N^{-1}y) e + (G^\dagger y^\dagger N^{-1}yG + y^\dagger G^{-1}y) a_{fg} = G^\dagger y^\dagger N^{-1}d$

Writing the matrix representation for this system gives

$$\begin{bmatrix} y^\dagger N^{-1}y + y^\dagger E^{-1}y & y^\dagger N^{-1}yG \\ G^\dagger y^\dagger N^{-1}y & G^\dagger y^\dagger N^{-1}yG + y^\dagger F^{-1}y \end{bmatrix} \begin{bmatrix} e \\ a_{fg} \end{bmatrix} = \begin{bmatrix} y^\dagger N^{-1}d + E^{-1/2}\omega_e + y^\dagger N^{-1/2}\omega_n \\ G^\dagger y^\dagger N^{-1}d + G^\dagger y^\dagger N^{-1/2}\omega_n \end{bmatrix}$$

▼ A previous version of this equation has looked like the following:

$$\begin{bmatrix} E^{-1} + y^\dagger N^{-1}y & y^\dagger N^{-1}yG \\ G^\dagger y^\dagger N^{-1}y & G^\dagger y^\dagger N^{-1}yG \end{bmatrix} \begin{bmatrix} e \\ a_{fg} \end{bmatrix} = \begin{bmatrix} y^\dagger N^{-1}d + E^{-1/2}\omega_e + y^\dagger N^{-1/2}\omega_n \\ G^\dagger y^\dagger N^{-1}d + G^\dagger y^\dagger N^{-1/2}\omega_n \end{bmatrix}$$

And the matrix representation for the systematics component is written as:

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To implement this in the code, this had to be realified using the formalism defined in [6 Realification Maths](#). The equations implemented in the code are described below (LHS and RHS separately):

LHS:

$$\begin{bmatrix} B^{-1} & 0 \\ 0 & B^{-1} \end{bmatrix} + \begin{bmatrix} H_{re}^T & H_{im}^T \\ -H_{im}^T & H_{re}^T \end{bmatrix} \begin{bmatrix} s_{re}^T & s_{im}^T \end{bmatrix} \begin{bmatrix} N^{-1} & 0 \\ 0 & N^{-1} \end{bmatrix} \begin{bmatrix} s_{re} \\ s_{im} \end{bmatrix} \begin{bmatrix} H_{re} & -H_{im} \\ H_{im} & H_{re} \end{bmatrix} \begin{bmatrix} b_{sy s_{re}} \\ b_{sy s_{im}} \end{bmatrix}$$

RHS:

RHS had to be expanded and implemented in the following form due to computational challenges:

$$\begin{bmatrix} H_{re}^T s_{re}^T N^{-1} d_{re} + H_{im}^T s_{im}^T N^{-1} d_{im} + H_{re}^T s_{re}^T N^{-1/2} \omega_{re} + H_{im}^T s_{im}^T N^{-1/2} \omega_{im} \\ -H_{im}^T s_{re}^T N^{-1} d_{re} + H_{re}^T s_{im}^T N^{-1} d_{im} - H_{im}^T s_{re}^T N^{-1/2} \omega_{re} + H_{re}^T s_{im}^T N^{-1/2} \omega_{im} \end{bmatrix}$$